

Public Finance

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**Fundamental Tax Reform and
Consumption Taxation (Ch. 25)**



Why fundamental Tax Reform?

There are three reasons for pursuing a fundamental tax reform:

1. To increase tax compliance
2. To make the tax code simpler
3. To improve tax efficiency



Improving Tax Compliance

Tax compliance: Efforts to reduce the evasion of taxes.

- To improve it, the government must reduce the amount of **tax evasion** (illegal nonpayment of taxation)

Tax avoidance: Legal action to reduce tax burden.



APPLICATION: Tax Evasion 1

The rich are not the only ones who evade taxes...

- For years, taxpayers wanting to claim tax exemptions for dependents were required only to fill in the names of their dependents on tax forms
- When the US 1986 Tax Reform Act required filers to list as well the Social Security numbers of dependents over the age of 5, 6 million suddenly disappeared from the tax rolls!
- Two years later, when the tax law required that the Social Security numbers of child care providers be listed for workers to claim their childcare tax credits, **2.6 million child care providers disappeared!**



APPLICATION: Tax Evasion 2

- Greek property taxes are based on self-reported house values, which led to widespread underreporting of home values. In a recent year, only 324 residents in a wealthy suburb reported having a pool.
- After former president George Papandreou used extreme means, such as helicopter surveillance and Google Earth pictures, there was a boom in the sales of tarpaulins to cover pools and make them invisible to aerial tax inspectors.



APPLICATION: Tax Evasion 3



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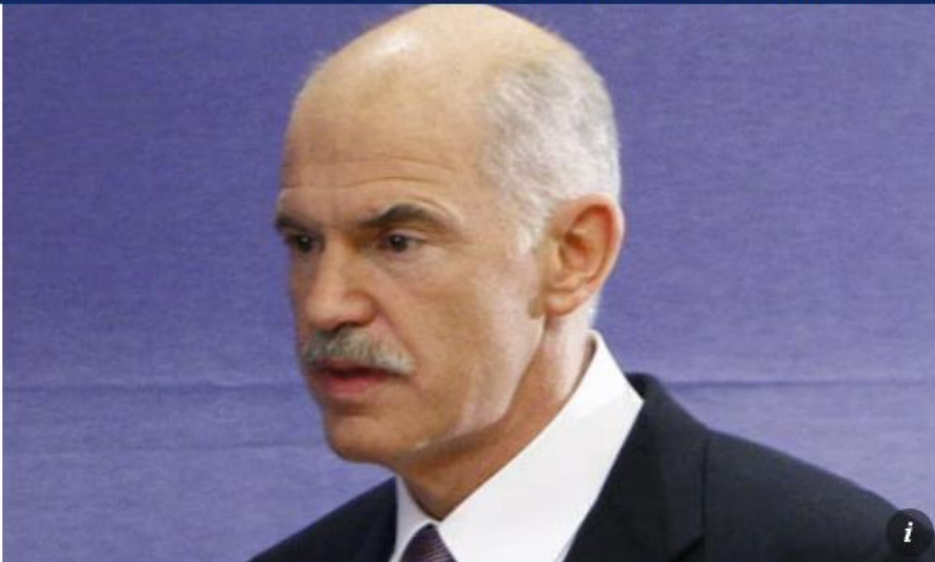
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Greece promises to crack down on tax evaders

Thursday, Nov. 27 2025

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Using data from the Cadaster, the Authority will cross-check information with tax returns and Form E9 to identify taxpayers who have not declared their swimming pools.

Theory of Tax Evasion

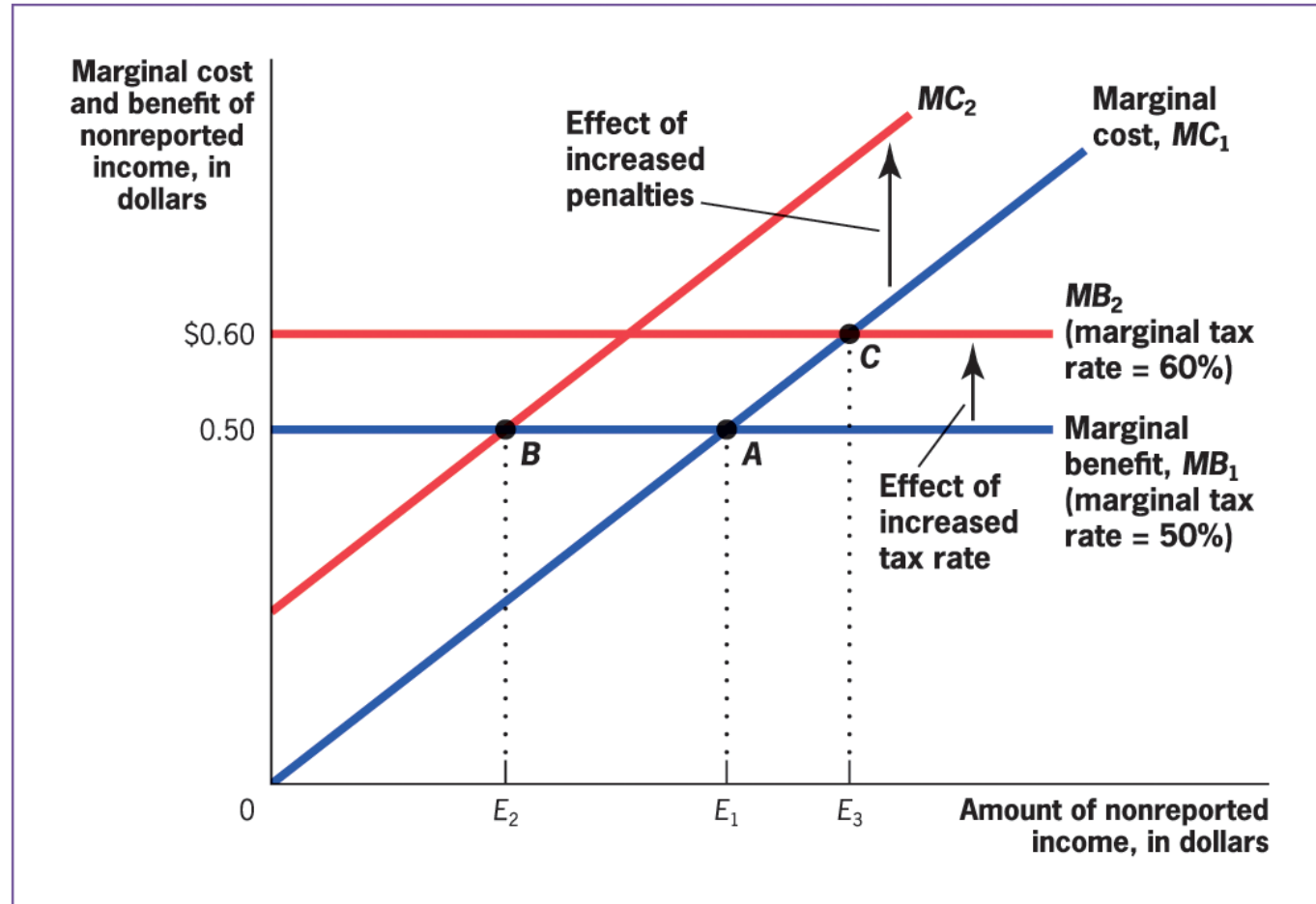
The theory of tax evasion emphasizes the trade-offs involved:

- Save money on unpaid taxes if not caught.
- Penalties or jail if caught.
- Higher taxes increase the benefit of avoidance without increasing the cost, increasing evasion.
- Increased penalties increase the cost of evasion and therefore reduce it.



Theory of Tax Evasion

Figure 25-1



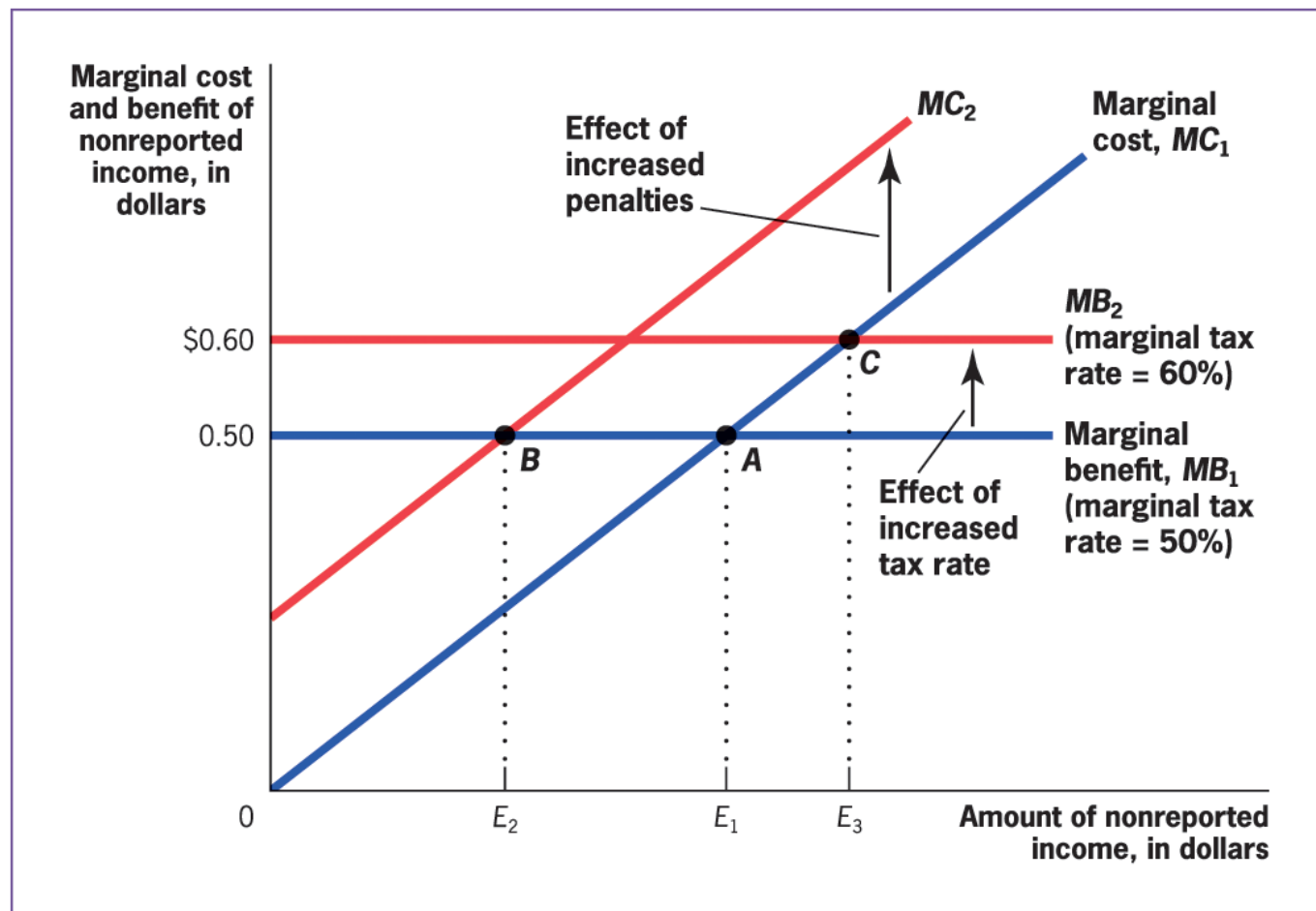
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The marginal benefit of evading taxes is the tax payment saved per dollar of evasion—the tax rate. Optimal evasion occurs when $MC = MB$, at E_1 .

Theory of Tax Evasion

Figure 25-1



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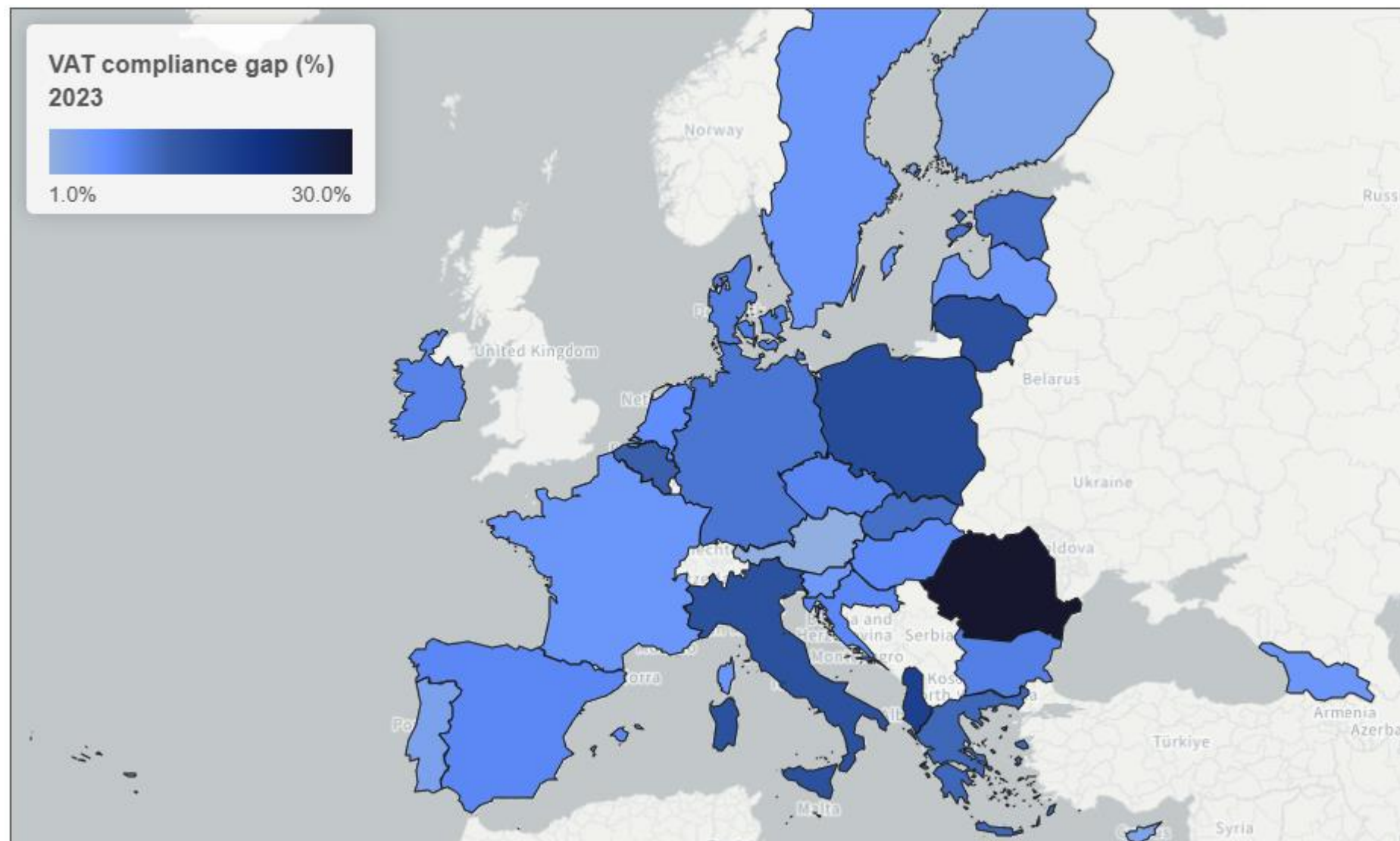
- If the penalties for the next dollar of cheating or the probability of getting caught rise $\rightarrow$ the marginal cost curve shifts up from MC_1 to MC_2 and the optimal amount of evasion is identified by point B. **INDIVIDUALS CHEAT LESS**
- If the marginal tax rate rises to 60% (holding penalties and thus marginal cost constant) $\rightarrow$ the marginal benefit curve shifts up from MB_1 to MB_2 . **INDIVIDUALS CHEAT MORE**

Evidence on Tax Evasion

- Tax evasion is pervasive in the United States and around the world.
 - U.S. “tax gap” between taxes owed and taxes paid was \$630 billion, 16.7% of tax revenue in 2020.
 - Sweden estimated its tax gap at 10% of tax revenue in 2008.
 - Worse in developing countries: 69% tax gap in Pakistan.
- Evidence that evasion increases with tax rates and decreases with threats of audits.



Evidence on Tax Evasion: Europe



Source: European Commission



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EMPIRICAL EVIDENCE: What Determines Tax Compliance?

Using a sample of 40,000 Danish taxpayers, Kleven et al. (2011) randomly selected half to be audited, whereas the remainder were not. The next year, a new randomization was done, with half receiving letters threatening an audit and half not.

Several findings:

- Evasion of income reported by third-party sources was quite low.
- Evasion of self-reported income was substantial and responded strongly to being audited or threatened with an audit.
- Higher tax rates led to more evasion but primarily had other effects, such as legal tax avoidance.



EMPIRICAL EVIDENCE: What Determines Tax Compliance?

- Slemrod et al. (2001) found that threatening an audit led to increased reported income for low- and middle-class families but lower reported income for upper-class families.
- Boning et al. (2018) found that auditing firms caused them to report more income, but it also caused firms that were served by the same tax preparer to report higher incomes, suggesting network effects of government enforcement.
- Slemrod et al. (2019) found that when Pakistan enacted a public disclosure program, forcing all taxpayers to reveal their reported tax liabilities, tax compliance rose disproportionately among those most likely to be identified by the disclosure—for example, politicians and people with unusual names.



Why should we care about Tax Evasion?

- Efficiency
 - Tax evasion narrows the base, reducing efficiency.
- Vertical equity
 - The wealthy can evade more easily, reducing their tax burden at the expense of poorer Americans.
- Especially: horizontal equity
 - A tax evader with the same income as a non-evader clearly has a lower tax burden.



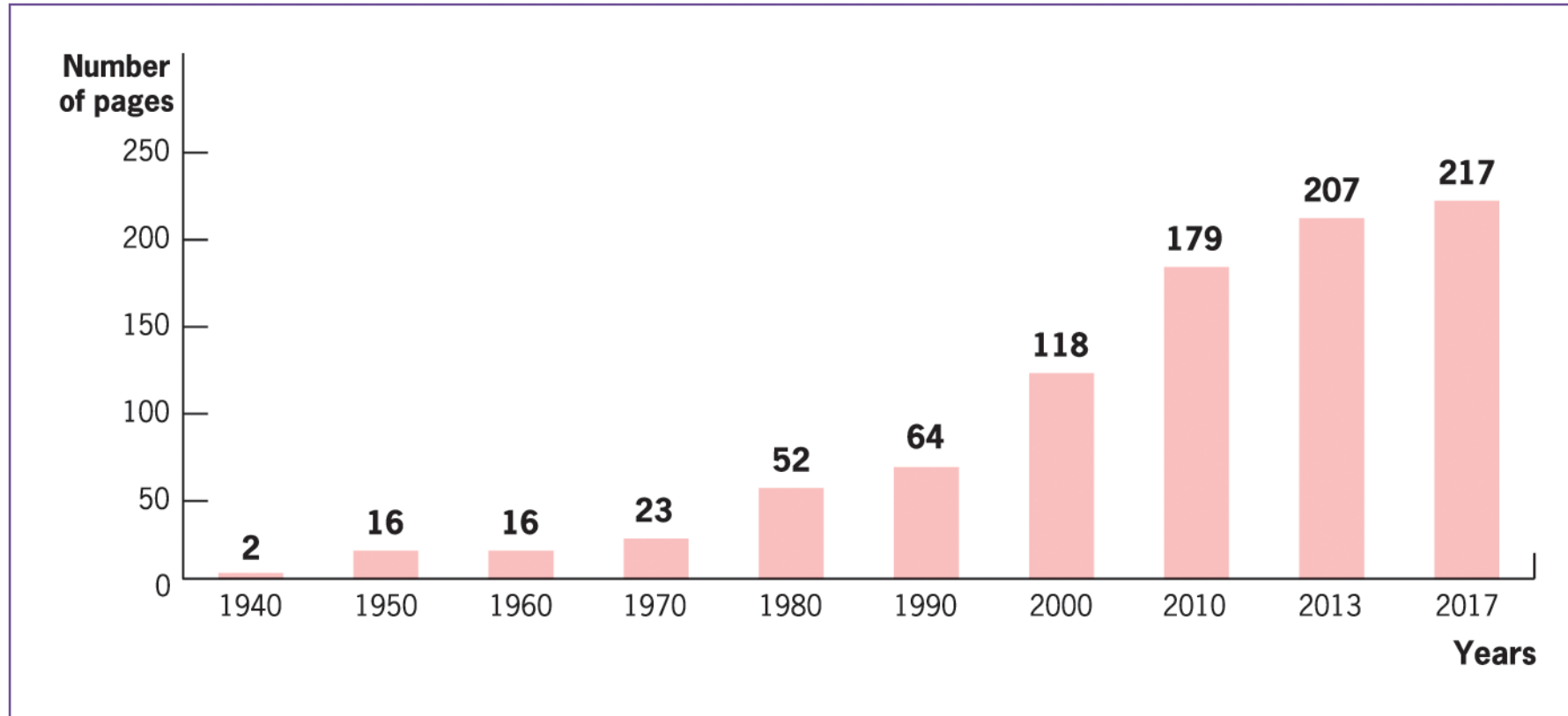
Making the Tax Code Simpler

- At the end of 2014, the IRS sent taxpaying individuals a packet with instructions for completing their Form 1040. The IRS estimated that it would take about **16 hours to complete the tax forms.**
- In 2000, taxpayers spent 3.2 billion hours and \$18.8 billion filling out tax forms, an average of 26.4 hours and \$209 per filer.
- Reducing exemptions or deductions would simplify the tax code.
 - But this may increase reporting requirements, with an unclear total impact on simplicity.



Making the Tax Code Simpler

Figure 25-2



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The **number of pages of instructions** that come with the basic individual tax form: the 1040.

This has risen from **2** pages in 1940 to **217** pages in 2017



Improving Tax Efficiency

- What matters for the efficiency cost of taxation are the **behavioral responses to taxation**
 - We can summarize these responses to taxation by asking:
How does changing the tax rate change tax revenues?
- Efficiency of the tax code depends on the **elasticity of revenues** with respect to the tax rate.
- Tax code changes have two effects on revenue:
 - Direct effect of tax changes: A higher tax rate that raises revenues on a fixed base of taxation.
 - Indirect effect of tax changes: A higher tax rate that decreases the revenue base on which taxes are levied.



Improving Tax Efficiency

Tax changes can have four **indirect effects**:

- Gross income effect: A higher tax rate may reduce gross income by lowering labor supply, savings, or risk taking.
- Reporting effect: For a given level of gross income, a higher tax rate causes people to report less income to avoid additional taxes.
- Income exclusion effect: For a given reported income, a higher tax rate causes people to take more deductions and exclusions.
- Compliance effect: Higher tax rates may reduce revenues through increased tax evasion.



Summary: The Benefits of Fundamental Tax Reform

Fundamental tax reform helps address all of the tax reform goals:

- Improves tax compliance and tax efficiency by expanding the tax base and lowering tax rates.
- Simplifies tax filing by ending many detailed exemptions and deductions and taxing different forms of income at the same rate.



The Politics and Economics of Tax Reform

- The Tax Reform Act (TRA) 86 broadened the base and reduced rates, representing a “victory” for tax reform.
- But this victory was short lived:
 - In 1993, top tax rates rose.
 - The 1997 Taxpayer Relief Act gave many new credits.
 - Tax reforms in 2001 and 2003 continued to complicate the tax code.
- Why is it so hard to maintain a simple, broad-based tax code?



Political Pressures for a Complicated Tax Code

- Political pressures are strongest when the winners are concentrated and have much to gain and the losers are diffused and don't lose much per person.
- Naïve voters may oppose explicit government spending but support tax expenditures.



Economic Pressures Against Broadening the Tax Base

- Any attempt to broaden the base may simply encourage additional tax shelters, undoing the base increase
- **Tax shelters:** Activities whose sole reason for existence is tax minimization
 - Tax shelters have real economic costs but save the shelters' creators money by reducing their tax burden



Economic Pressures Against Broadening the Tax Base

- BACKGROUND: by the mid-1980s, the tax code had created a number of ways in which savvy inventors could legally turn a profit by shielding their money from the IRS
 - Many measures were intended by the Congress to encourage investment in real estate and oil industry, but the distortion on incentives created serious overinvestments
- **Tax shelters** run in opposition to the benefits of fundamental tax reform:
 - Tax shelters make tax evasion easier
 - They make the tax code more complicated
 - They make the tax code less efficient



Transitional Inequities 1

- Because of tax capitalization, eliminating tax shelters can create large transitional inequities
 - Tax capitalization: The change in asset prices that occurs due to a change in the tax levied on the stream of returns from that asset
 - Transitional inequities from tax reform: Changes in the treatment of similar individuals who have made different decisions in the past and are therefore differentially treated by tax reform



Transitional Inequities 2

- Transitional inequities are a natural feature of any tax reform, which by definition will create winners and losers
- Concerns addressing transitional inequities have led to various forms of compensation through the political process, such as **grandfathering**:
 - Those who made decisions under the old tax rules are allowed to continue to benefit from those rules, while the rules are changed for all future decisions



The Conundrum

- Political and economic pressures are significant barriers to moving to a broad-based system
- Political forces are constantly pushing for the use of the tax code to deliver benefits to particular groups at the cost of potentially inefficient and inequitable holes in the tax base
- After these tax breaks are in place, it is very hard to remove them because doing so would create horizontal inequities for those who made decisions based on the existence of these tax breaks



Consumption Taxation

- Consumption taxation is a radical reform preferred by many economists
- **Taxing consumption:** Taxing individuals based not on what they earn but what they consume, such as through a sales tax
- Used by state and local governments and many governments around the world



Why might consumption make a better Tax Base? 1

Potential advantages to moving to a consumption tax base:

- Improved capital allocation
- Fairer treatment of savers and less distortion to savings decision
- Simplicity



Why might consumption make a better Tax Base? 2

1. Improved Capital Allocation

- The current tax system encourages some kinds of investment over others
- Better allocation improves economic efficiency.



Why might consumption make a better Tax Base? 3

2. Fairer Treatment of Savers and Less Distortion to Savings Decisions

	Homer	Marge
Income Tax		
Income in period 1	\$100	\$100.00
Taxes in period 1	50	50.00
Consumption in period 1	50	25.61
Savings in period 1	0	24.39
Interest earnings in period 2	0	2.44
Taxes in period 2	0	1.22
Consumption in period 2	0	25.61
PDV of taxes	50	51.11
Consumption Tax		
Income in period 1	\$100	\$100.00
Consumption in period 1	50	26.19
Taxes in period 1	50	26.19
Savings in period 1	0	47.62
Interest earnings in period 2	0	4.76
Consumption in period 2	0	26.19
Taxes in period 2	0	26.19
PDV of taxes	50	50.00



Why might consumption make a better Tax Base? 4

3. Simplicity

- In principle, it is much more straightforward to simply tax individuals on their purchases than on a complicated definition of income.



Why Might Consumption Be a Worse Tax Base? 1

1. Vertical equity
2. Differences between savers and Non-savers
3. Transition issues
4. Compliance
5. Cascading



Why Might Consumption Be a Worse Tax Base? 2

1. Vertical equity

- Consumption taxes reduce vertical equity
- Rich people save more, so consumption taxes are regressive
- Net lifetime savings are bequests, so an estate tax might help, but estate taxes are very unpopular
- A progressive expenditure tax could correct this but would introduce other complications



Why Might Consumption Be a Worse Tax Base? 3

2. Differences Between Savers and Non-Savers

- Taxing savings might help tax high-ability people.

3. Transition Issues

- Seniors could be devastated by the transition.

4. Compliance

- It is harder to measure consumption than income.

5. Cascading

- One business's output is another's input, making it difficult to avoid double taxation



Designing a Consumption Tax: Value-Added Tax

- A value-added tax addresses cascading and compliance.
 - **Value-added tax (VAT):** A consumption tax levied on each stage of a good's production on the increase in value of the good at that stage of production.
- VATs are widely used around the world.
- Firms have an incentive to make sure that the firms they are buying from and selling to are honest, as dishonesty to reduce one firm's taxes would increase the other firms' taxes.



Value-Added Tax in Practice

Agent	Purchase Price	Sale Price	Value Added	Tax Paid (VAT = 20%)
Logger	\$0	\$25	\$25	\$5
Manufacturer	25	75	50	10
Retailer	75	100	25	5
			Total tax paid:	\$20

When the logger adds \$25 in value through producing the wood, **they pay** \$5 in VAT.

The manufacturer then adds \$50 in value and pays a VAT of \$10.

Finally, the retailer pays \$5 in VAT on her \$25 in value added.



Designing a Consumption Tax: Value-Added Tax

- The VAT is not that simple as it seems
- In practice, these VATs typically have multiple rates
 - Some lower rates for progressivity
 - Some lower rates for politics
 - Administrative costs are similar to income tax systems



Designing a Consumption Tax: Expenditure Tax

- An alternative to a VAT is an expenditure tax.
- **Expenditure tax:** A consumption tax levied on **yearly consumption** rather than on specific sales.
 - An expenditure tax operates in the same way as an income tax, but it replaces the base of taxation with expenditure rather than income
- It is straightforward to make an expenditure tax progressive, making the consumption tax system more vertically equitable.
- However, it is **difficult to track** expenditures throughout the year.



Backing into Consumption Taxation: Cash-Flow Taxation

- Previous issues:
 - Taxing consumption at the point of production leads to progressivity concerns
 - Taxing expenditure at the individual level is administratively infeasible→ Cash-flow taxation is a third alternative
- **Cash-flow tax**: A tax on the difference between cash income and savings
- Effectively taxes consumption and can be based on year-end total consumption

VAT in Italy (Imposta sul Valore Aggiunto)

- The standard VAT rate in Italy is **22%** and applies to most goods and services.
- Several reduced VAT rates apply to specific goods and services:
 - 4%** → goods of primary necessity (e.g., basic food products, books, newspapers)
 - 5%** → some food products and health or social services
 - 10%** → tourism, restaurants, transport, and some utilities
- Some activities are exempt (e.g., financial services, education, healthcare).
- Reduced rates reflect distributional and political considerations, aiming to lower the tax burden on essential goods and key sectors.

EXERCISES



MC 1

What is the primary economic motivation for fundamental tax reform?

- a) To increase statutory tax rates
- b) To improve efficiency and reduce compliance costs
- c) To reduce redistribution
- d) To shift taxation toward capital income



MC 2

Why does a more complex tax system tend to increase tax avoidance and evasion?

- a) Because complexity creates loopholes and grey areas
- b) Because enforcement agencies become weaker
- c) Because higher tax rates always induce evasion
- d) Because taxpayers face lower penalties



MC 3

In the standard economic model of tax evasion, an individual chooses how much to evade by comparing:

- a) Average tax rates and penalties
- b) Total tax liability and income
- c) Audit probabilities and government revenue needs
- d) Marginal benefits of evasion and marginal costs of evasion



MC 4

Why can base broadening improve both tax efficiency and revenue collection?

- a) It raises marginal tax rates
- b) It reduces avoidance opportunities and distortions
- c) It targets high-income taxpayers more heavily
- d) It eliminates the need for audits



MC 5

From an efficiency perspective, why is consumption taxation often preferred to income taxation?

- a) It increases tax progressivity
- b) It reduces government revenue volatility
- c) It taxes capital income more heavily
- d) It avoids taxing saving and intertemporal choices



True/False (please provide a justification)

- 1) In practice, increasing penalties for tax evasion always reduces the amount of evasion.
- 2) A value-added tax is generally less susceptible to evasion than a retail sales tax.
- 3) Base broadening can allow governments to lower marginal tax rates without reducing total tax revenue.



EXERCISE 1

An individual earns €50000 and faces an income tax rate of 30%. The individual considers evading €10000 of income.

- Probability of audit: 10%
- Penalty rate: 200% of unpaid tax
 - a) Calculate the tax saved if evasion is not detected.
 - b) Calculate the expected penalty associated with evasion.
 - c) Determine whether evasion is optimal in expected value terms and explain briefly.



EXERCISE 2

An individual earns €50000 in period 1. They consume €35000 in period 1 and save the remainder. Savings earn a 10% return, which is entirely consumed in period 2.

The tax rate is 30% under both systems.

- a) Compute total lifetime tax paid under an income tax.
- b) Compute total lifetime tax paid under a consumption tax.
- c) Compare the two outcomes and explain which tax system distorts intertemporal choices less and why.

